



Single Sign-On Implementation Guide

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tl;dr Summary	You can set up a SAML SSO integration for both users and consumers. You can also set up an SSO for both your production and demo environments. In this setup, Eltropy acts as the service provider (SP), and you must provide your own third-party identity provider (IdP) or use your own in-house IdP method. Eltropy only supports IdP-initiated SAML SSO. The Support Center uses email addresses to connect rep and consumer accounts with your IdP. You can also use a unique External ID for your rep accounts if their email addresses do not match up between your systems. Eltropy offers two API versions. One passes credentials with SAML attributes and the other uses SAML NameID subject. Your IdP provider may dictate which version you can use. For consumer SSO, the system uses the first_name and last_name attributes for auto-provisioning. The “Representative SSO” and “Consumer SSO” sections in this document provide the necessary information that you must enter when configuring your IdP, like the relay state URL and issuer ID. Reference the “Preparatory Requirements” section to ensure that your plan to implement SSO includes all the necessary components. Lastly, you must work with your Eltropy CSM to implement an SSO. Your CSM must enable certain settings in your Support Center, plus, you must also supply your CSM with your IdP metadata.
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Introduction

Single sign-on (SSO) lets your users access your Support Center without needing to log in every time. Instead, they can launch the Support Center from an identity provider (IdP) platform. You can use this IdP to also launch other apps that users often use, like email or messaging apps. An IdP simplifies your users' work and increases security; users aren't required to manage unique account credentials for all the apps they use.

You can also set up an SSO for your consumers. This lets the consumer go directly from your financial portal into an Eltropy call without entering any credentials. Instead, Eltropy uses their basic account information from the financial portal like name and phone number.

Eltropy provides API endpoints for a security assertion markup language (SAML) SSO. SAML provides a secure way of sharing account credentials to create the SSO functionality. You have options for two different visions, depending on how you want to share credentials and which IdP you use. These two endpoint versions have small differences, which are explained in the "V1 or V2" section of this document.

The simplest method of setting up an SSO is to use a third-party IdP. However, you can also use an in-house SSO setup that your organization created. This document focuses on implementing SSO using a third-party IdP for users and using your consumer's financial portal credentials for your Client Portal. If you want to use an in-house SSO, this guide may be helpful, but you may have additional requirements. Contact your CSM for assistance.

Tip:

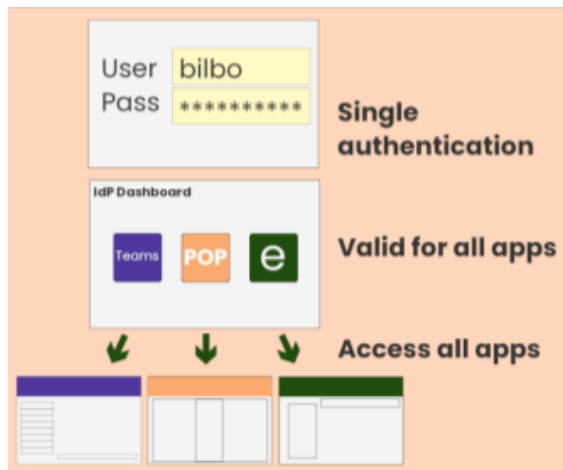
You can implement SSO for both your production and demo systems.

What Your Users See

The exact experience that your users see may change based on your exact setup, like which IdP you use. The following are two generic examples of what a user and consumer might experience with a standard setup:

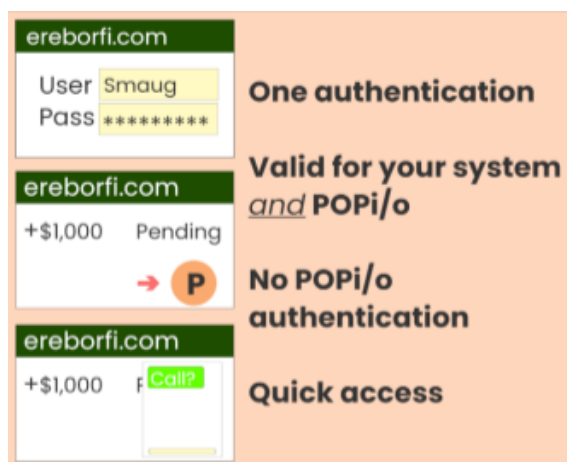
User example:

You have Okta set up as your IdP to manage your users' different accounts. Bilbo the rep uses an internal messaging application, an ACD application, and also the Support Center. Instead of opening three different apps and logging into each one, Bilbo only signs in to Okta. From the Okta dashboard, Bilbo quickly opens all three applications. He's automatically logged in when he opens each of the apps because of the SSO.



Consumer example:

A consumer logs in to your financial portal to verify their account balance. They deposited a check with one of your reps and have a question about their available funds. So, they click a button on your portal to "Talk with an agent". This re-directs the consumer to your Eltropy Client Portal. Since they already logged in to the financial portal, the same credentials are used for the Client Portal. On the Client Portal, they're already logged in and can immediately select a team or rep to call. The SSO streamlined the experience, removing the requirements to authenticate a second time before starting a call.



How It Works

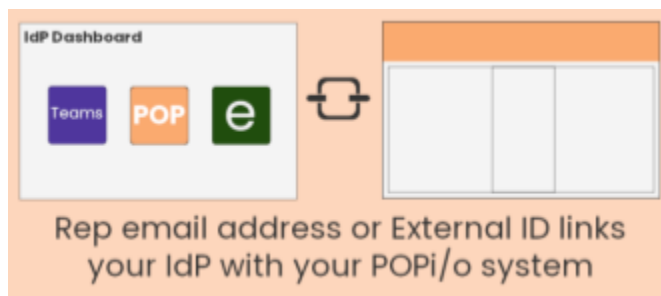
The following are high-level technical descriptions of how user and consumer SSO work.

User SSO:

You set up your users in a third-party identity provider (IdP), such as Okta or Centrify. You map the user's Support Center user accounts to the IdP with their email address. Map the email using a SAML [NameID](#) subject or a SAML attribute. The two Eltropy endpoint versions allow you to use either the attributes or the subject. See the "V1 or V2" section of this document to assess which version is best for you. After configuring your IdP, it will send a metadata XML snippet with your user's IdP credentials when the user launches the Support Center from the IdP.

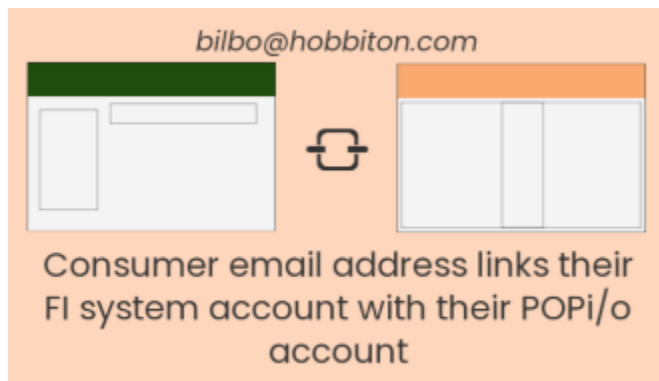
Tip:

Instead of email, you can optionally use a unique identifier instead. If you want to use a unique identifier, you can add one to each Support Center user account in your Employee Directory. This is further explained in the "Preparatory Requirements" section.



Consumer SSO:

Consumer SSO uses the consumer's email address as the connection between the Client Portal and your financial system. When they launch the Client Portal to start a call, the email address for their financial portal account links with their Client Portal account email address. Depending on your requirements and preference, the consumer email is passed in the SAML metadata as either a SAML attribute or SAML subject. If their email does not exist in your Client Portal, the system looks for the [first_name](#) and [last_name](#) SAML attributes. If these two attributes are configured, a new consumer account is automatically created. You can access consumer accounts in your Consumer Directory (menu > **Administration Pages** > **Consumer Directory**). See the "V1 or V2" section of this document to assess which endpoint version is best for you.



The following is an example SAML assertion:

```
<?xml version="1.0" encoding="UTF-8"?>
<saml2:Assertion ID="id76727272639007758675309" IssueInstant="2021-11-16T17:39:11.107Z"
Version="2.0" xmlns:saml2="urn:oasis:names:tc:SAML:2.0:assertion">
  <saml2:Issuer
Format="urn:oasis:names:tc:SAML:2.0:nameid-format:entity">POPIO_REP_SAML_ISSUER_erebor</saml
2:Issuer>
  <saml2:Subject>
    <saml2:NameID
Format="urn:oasis:names:tc:SAML:1.1:nameid-format:unspecified">thorinoakenshield@ereborfi.co
m</saml2:NameID>
    <saml2:SubjectConfirmation Method="urn:oasis:names:tc:SAML:2.0:cm:bearer">
      <saml2:SubjectConfirmationData NotOnOrAfter="2021-11-16T17:44:11.107Z"
Recipient="https://main.example.com/simplesaml/module.php/{removed URL segment}/sso"/>
    </saml2:SubjectConfirmation>
  </saml2:Subject>
  <saml2:Conditions NotBefore="2021-11-16T17:34:11.107Z"
NotOnOrAfter="2021-11-16T17:44:11.107Z">
    <saml2:AudienceRestriction>
      <saml2:Audience>sso</saml2:Audience>
    </saml2:AudienceRestriction>
  </saml2:Conditions>
  <saml2:AuthnStatement AuthnInstant="2021-11-16T17:39:11.107Z"
SessionIndex="id1637084351105.8675309">
    <saml2:AuthnContext>
      <saml2:AuthnContextClassRef>urn:oasis:names:tc:SAML:2.0:ac:classe
s:PasswordProtectedTransport</saml2:AuthnContextClassRef>
    </saml2:AuthnContext>
  </saml2:AuthnStatement>
</saml2:Assertion>
```

V1 or V2

Eltropy provides two API versions for SSO. They have two main differences:

- Both versions use a separate ACS URL. These two URL versions are provided in the setup instructions, later in this document.
- The versions pass user credentials differently. This may affect which version you use since your IdP may only support one method.

- V2 uses NameID SAML subject

- V1 uses SAML attributes:

Attribute	Details	Required?
fi_id	An internal ID from the Support Center that identifies your organization.	Depends on your SAML configuration.
uid	User ID, provided by the IdP, that uniquely identifies the member.	Yes
email	The user's email. This must be unique within the Support Center.	Only if provisioning a new user.
first_name	User's first name.	Only if provisioning a new user.
last_name	User's surname.	Only if provisioning a new user.

Preparatory Requirements

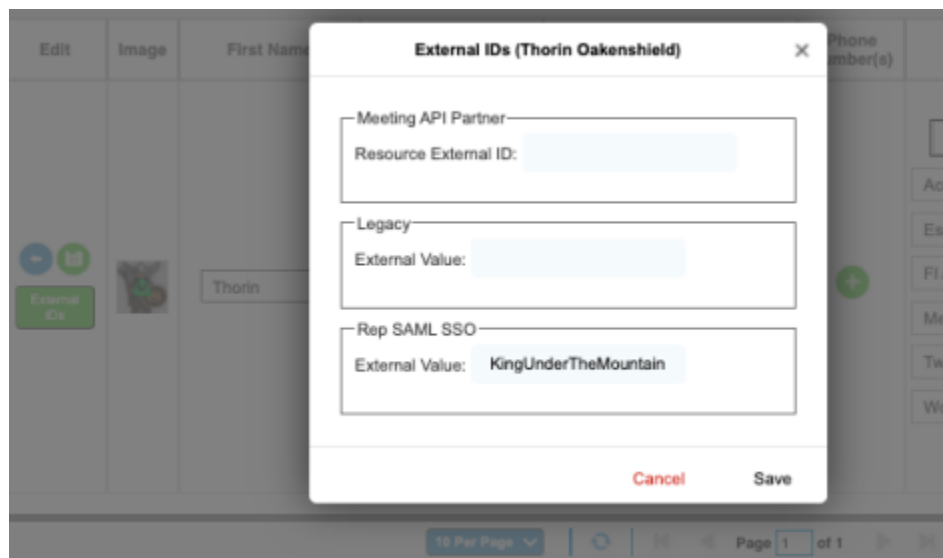
Before beginning your SSO setup, complete the following items:

- Identify your Eltropy Support Center company code. You can find this at the end of your Support Center URL. For example, the end of the URL for the Erebor FI is “...support_center&company_code=erebor”. Contact your CSM if you have any questions.
- For user SSO, set up an existing third-party IdP for your organization, such as Okta or SailPoint.
- Each user with access to the Support Center must have an IdP user account. If you prefer to use an in-house SSO, contact your CSM to coordinate customization requirements.
- Your CSM must enable the necessary SSO settings in your Support Center. When doing so, the CSM also determines if you want to use email addresses or a unique ID to link the user accounts between your IdP and the Support Center (see next bullet point).
- Assess whether you will use the V1 or V2 endpoint. Differences are explained in the “V1 or V2” section earlier in this document.
- Ensure you do not have other namespaces in the user attributes or group attributes.
- Plan how you will link accounts between systems:
 - **For consumer SSO:**
Map the consumer email address between your financial system and your Client Portal.
 - **For user SSO:**
You have three options for mapping accounts between your IdP and Support Center.
 1. The default method is to match email addresses. You must ensure that each user’s email is the exact same in both your Support Center and IdP.

2. Use unique identifiers by manually inserting External Value IDs for each user's Support Center account. You can do so when editing an account in the Employee Directory. Enter the ID in the [Rep SAML SSO](#) field.

Important!

You may use any format for your unique IDs. The only requirement is that the first character to be alphanumeric.



3. Use SAML attributes to link your user's email address between your IdP and your Support Center. These are typically set up during the configuration process in your IdP. If you use this method, the user's email address must match between the two apps. Use [email](#) as the attribute name.

Note:

Depending on your IdP, like Microsoft Azure, this option may be your only available method.

Set Up User SSO

The exact steps to set up SSO vary depending on your choice of identity provider (IdP). For example, Okta requires you to create a new app integration with the type of SAML 2.0 inside of Okta. Then, you enter the necessary information to auto-generate settings of the new app integration. Since each SSO setup is different, the following are generic steps for a standard SSO setup. These steps include the required information for a setup, but you must work with your CSM if you have questions specific to your IdP.

1. Be sure to complete the “Preparatory Requirements” section of this document.
2. Configure the necessary items in your IdP. This may entail setting up an application, creating a hosted identity provider, or similar; it depends on your choice of IdP.

The following items are information from Eltropy that you must enter into your IdP:

- Enter one of the ACS URLs listed below. Your IdP may refer to this URL as the Sign-On URL. This is the endpoint to which your IdP will direct users, along with its signed /base-64 encoded SAMLResponse parameter.
 - i. **V1 endpoint:**
`https://main.financialtown.com/simplesaml/module.php/{removed URL segment}/member_<company_code>`
 - ii. **V2 endpoint:**
`https://main.financialtown.com/financial_town/?view_type={removed URL segment}rep_sign_on&company_code=<company_code>`
- Enter `POPIO_REP_SAML_ISSUER_<company_code>` as your issuer or entity ID. Be sure to replace `<company_code>` with your actual company code.

Tip:

In the metadata, this field is a property of the ENTITYDescriptor tag called the entityID. When you send the assertion it is the value inside the issuer tag (see the example metadata assertion at the end of this task).

-
- Enter **SSO** for your audience.
 - If you're using SAML attributes, add **email** as the attribute name. The attribute value is the user's email address. Remember that the email addresses must match in both your IdP and Support Center.
 - Enter the following URL for the Relay State (the URL to which your reps will be redirected upon successful authentication):
https://main.financialtown.com/financial_town/{removed URL segment}rep_sign_on&company_code=<company_code>
3. Complete any other configuration steps that your specific IdP may require.

Important!

After configuring the above items in your IdP, part of the process (depending on your IdP) may include generating a metadata code snippet. Your IdP may generate the actual snippet, or it may provide you with a URL from which you can access the metadata.

4. Generate a metadata snippet or URL.
5. Send your metadata snippet or the metadata URL to your CSM. Your CSM uses the metadata to complete the setup.

Example metadata URL:

8675309.okta.com/app/0513DEBE2263D4885A9F174D2155A106/sso/saml/metadata

Example metadata snippet:

```
<?xml version="1.0"?>
<md:EntityDescriptor xmlns:md="urn:oasis:names:tc:SAML:2.0:metadata"
entityID="POPIO_REP_SAML_ISSUER_erebor">
<md:IDPSSODescriptor WantAuthnRequestsSigned="false"
protocolSupportEnumeration="urn:oasis:names:tc:SAML:2.0:protocol">
<md:KeyDescriptor use="signing">
<ds:KeyInfo xmlns:ds="http://www.w3.org/2000/09/xmldsig#">
<ds:X509Data>
<ds:X509Certificate>MIIDq...iUEisow4pIw==</ds:X509Certificate>
</ds:X509Data>
```

```
</ds:KeyInfo>
</md:KeyDescriptor>

<md:NameIDFormat>urn:oasis:names:tc:SAML:1.1:nameid-
format:unspecified</md:NameIDFormat>
<md:NameIDFormat>urn:oasis:names:tc:SAML:1.1:nameid-
format:emailAddress</md:NameIDFormat>

<md:SingleSignOnService
Binding="urn:oasis:names:tc:SAML:2.0:bindings:HTTP-POST"

Location="https://8675309.okta.com/app/8675309_popiodemoereborrepssso
_1/exk2lbcewh9vCP9e05d7/sso/saml"/>

<md:SingleSignOnService
Binding="urn:oasis:names:tc:SAML:2.0:bindings:HTTP-Redirect"

Location="https://8675309.okta.com/app/dev-
13331348_popiodemoereborrepssso_1/exk2lbcewh9vCP9e05d7/sso/saml"/>

</md:IDPSSODescriptor>
</md:EntityDescriptor>
```

Set Up Consumer SSO

A typical SSO experience for consumers does not include a third-party identity provider (IdP). Instead, they typically use your own system as the IdP. Therefore, this section focuses on the use-case of setting up an SSO with your own in-house IdP. Since every setup is different, your team must use the provided information and integrate it into your system however they see fit.

1. Be sure to complete the “Preparatory Requirements” section of this document.
2. Configure your IdP settings with the following information:
 - a. If you are configuring the V2 endpoint, enter the following ACS URL. This URL may also be known as the Sign-On URL. This is the endpoint to which your IdP will direct consumers, along with its signed/base-64 encoded SAMLResponse parameter.
<https://main.financialtown.com/{removed URL segment}/sso>
 - b. Enter `POPIO_REP_SAML_ISSUER_<company_code>` as your issuer or entity ID. Be sure to replace `<company_code>` with your actual company code.

Tip:

In the metadata, this field is a property of the ENTityDescriptor tag called the entityID. When you send the assertion it is the value inside the issuer tag (see the example metadata assertion at the end of this task).

- c. Enter `SSO` for your audience.
 - d. Enter the following URL for the Relay State (the URL to which your reps will be redirected upon successful authentication):
https://main.financialtown.com/financial_town/{removed URL segment}rep_sign_on&company_code=<company_code>
3. Configure consumer auto-provisioning:
Auto-provisioning is when the Support Center automatically creates a consumer account. This happens when the Support Center does not recognize the email that is

passed with the **NameID** in the SAML assertion. When this happens, a new consumer account is created with the email and the consumer is automatically logged in (and can start a call). For this to happen, your video banking system will look for two attributes in the assertion: **first_name** and **last_name**. Be sure to include these two attributes in your SAML assertions.

4. Finish configuring any other features of your SAML metadata snippet.
5. Send your SAML metadata snippet to your Eltropy CSM. Your CSM uses the metadata to complete the setup.

Example metadata snippet:

```
<?xml version="1.0"?>
<md:EntityDescriptor xmlns:md="urn:oasis:names:tc:SAML:2.0:metadata"
entityID="POPio_MEMBER_SAML_ISSUER_erebor">
<md:IDPSSODescriptor WantAuthnRequestsSigned="false"
protocolSupportEnumeration="urn:oasis:names:tc:SAML:2.0:protocol">
<md:KeyDescriptor use="signing">
<ds:KeyInfo xmlns:ds="http://www.w3.org/2000/09/xmldsig#">
<ds:X509Data>
<ds:X509Certificate>MIIDqD...+EiGARNiUEisoW4pIw==</ds:X509Certificate
>
</ds:X509Data>
</ds:KeyInfo>
</md:KeyDescriptor>

<md:NameIDFormat>urn:oasis:names:tc:SAML:1.1:nameid-
format:unspecified</md:NameIDFormat>

<md:NameIDFormat>urn:oasis:names:tc:SAML:1.1:nameid-
format:emailAddress</md:NameIDFormat>

<md:SingleSignOnService
Binding="urn:oasis:names:tc:SAML:2.0:bindings:HTTP-POST"

Location="https://8675309.okta.com/app/8675309_ereborfi_1/exk1z2zxzrt
72qn0h5d7/sso/saml"/>
```



```

<md:SingleSignOnService
Binding="urn:oasis:names:tc:SAML:2.0:bindings:HTTP-Redirect"

Location="https://8675309.okta.com/app/dev-
13331348_ereborfi_1/exk1z2zxzrt72qn0h5d7/sso/saml"/>

</md:IDPSSODescriptor>
</md:EntityDescriptor>

```

Example SAML assertion:

```

<?xml version="1.0" encoding="UTF-8"?>
<saml2:Assertion ID="id79387981530640151538387607"
IssueInstant="2021-11-
19T18:49:25.985Z" Version="2.0"
xmlns:saml2="urn:oasis:names:tc:SAML:2.0:assertion">

<saml2:Issuer Format="urn:oasis:names:tc:SAML:2.0:nameid-
format:entity">POPIO_MEMBER_SAML_ISSUER_erebor</saml2:Issuer>

<saml2:Subject>
<saml2:NameID
Format="urn:oasis:names:tc:SAML:1.1:nameid-format:unspecified">bilbo@
hobbiton.com</saml2:NameID>
<saml2:SubjectConfirmation
Method="urn:oasis:names:tc:SAML:2.0:cm:bearer">
<saml2:SubjectConfirmationData
NotOnOrAfter="2021-11-19T18:54:25.985Z"
Recipient="https://main.financialtown.com/{removed URL segment}/saml2-
acs.php/sso"/>
</saml2:SubjectConfirmation>
</saml2:Subject>
<saml2:Conditions NotBefore="2021-11-19T18:44:25.985Z"
NotOnOrAfter="2021-11-
19T18:54:25.985Z">
<saml2:AudienceRestriction>
<saml2:Audience>sso</saml2:Audience>

```

```
</saml2:AudienceRestriction>
</saml2:Conditions>
<saml2:AuthnStatement AuthnInstant="2021-11-19T18:49:25.985Z"
SessionIndex="id1637347765984.8675309">
<saml2:AuthnContext>
<saml2:AuthnContextClassRef>urn:oasis:names:tc:SAML:2.0:ac:classes:Pa
sswordProte
ctedTransport</saml2:AuthnContextClassRef>
</saml2:AuthnContext>
</saml2:AuthnStatement>
<saml2:AttributeStatement>
<saml2:Attribute Name="first_name"
NameFormat="urn:oasis:names:tc:SAML:2.0:attrname-format:unspecified">
<saml2:AttributeValue
xmlns:xs="http://www.w3.org/2001/XMLSchema"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:type="xs:string">Bilbo
</saml2:AttributeValue>
</saml2:Attribute>
<saml2:Attribute Name="last_name"
NameFormat="urn:oasis:names:tc:SAML:2.0:attrname-format:unspecified">
<saml2:AttributeValue
xmlns:xs="http://www.w3.org/2001/XMLSchema"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:type="xs:string">Baggins
</saml2:AttributeValue>
</saml2:Attribute>
</saml2:AttributeStatement>
</saml2:Assertion>
```